

The hook rank formula for the staircase Bott–Samelson on $V_{(n-1,1)}$: structural proof of the first R'_n block, plus inductive program

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Abstract

For the staircase Bott–Samelson element $\Pi^{S_n} = \prod_{k=1}^{n-1} (T_k+1) \cdots (T_1+1)$ in the Iwahori–Hecke algebra $H_q(S_n)$, computational data through $n \leq 8$ supports

$$\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \left\lceil \frac{n-2}{2} \right\rceil \quad (n \geq 2).$$

We give a rigorous structural proof of the first iterative-image step (Lemma 5): the image of $V_{(n-1,1)}$ under the first block $R'_n = (T_{n-1}+1) \cdots (T_1+1)$ is $\text{span}(v_2, v_3, \dots, v_{n-2}, u_{n-1})$, of dimension $n-2$. For the subsequent blocks, we set up an inductive image-tracking program and verify it computationally for $n \leq 8$. The conjectural “free v_2 in Phase 1, tied v_2 in Phase 2” transition — which is the heart of the rank formula — is supported by all data and a sketch via branching, but a fully rigorous proof of the transition is deferred. We conclude with the explicit dichotomy for the kernel of $\Pi^{S_m}|_{V_{(m-1,1)}}$ that an inductive proof would require.

1 Setup

1.1 Conventions

Let $H_q(S_n)$ be the Iwahori–Hecke algebra of type A_{n-1} with generators $T_1, \dots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid relations. We work over the field $\mathbb{Q}(q)$, so all Hoefsmit denominators ($1 - q^d$ for axial distance $d \neq 0$) are invertible.

1.2 The standard module $V_{(n-1,1)}$

The cell module $V_{(n-1,1)}$ has dimension $n-1$, indexed by the standard Young tableaux (SYTs) of shape $(n-1, 1)$. The SYT $T^{(k)}$ ($k = 2, \dots, n$) has entry k in row 2 at $(2, 1)$ and the remaining entries $\{1, \dots, n\} \setminus \{k\}$ in row 1 in increasing order. We write $v_k := v_{T^{(k)}}$ for the seminormal basis vector. So

$$V_{(n-1,1)} = \bigoplus_{k=2}^n \mathbb{Q}(q) v_k.$$

1.3 Action of T_i on $V_{(n-1,1)}$

Lemma 1 (Eigenstructure of T_1). $T_1 v_2 = -v_2$ and $T_1 v_k = q v_k$ for $3 \leq k \leq n$.

Proof. For $k = 2$: entries 1 and 2 both in column 1, so axial distance -1 , $T_1 v_2 = -v_2$. For $k \geq 3$: 1 at $(1, 1)$, 2 at $(1, 2)$ in $T^{(k)}$, axial distance 1, $T_1 v_k = q v_k$. $\square$

Corollary 2. $E_1^- = \mathbb{Q}(q) \cdot v_2$, $E_1^+ = \text{span}(v_3, \dots, v_n)$.

Lemma 3 (Block structure of T_i , $i \geq 2$). *For $i \geq 2$, on $V_{(n-1,1)}$:*

- $T_i v_k = q v_k$ for $k \notin \{i, i+1\}$.
- T_i acts as a 2×2 block on $\text{span}(v_i, v_{i+1})$ at axial distance $\pm i$, with eigenvalues q and -1 .

We write $u_i \in \text{span}(v_i, v_{i+1})$ for the $+q$ -eigenvector of T_i ($i \geq 2$). Then $u_i = \alpha_i v_i + \beta_i v_{i+1}$ for some $\alpha_i, \beta_i \in \mathbb{Q}(q)^\times$ depending only on the axial distance $\pm i$. Moreover:

Lemma 4 ((T_i+1) on the block). *For $i \geq 2$, $(T_i+1)v_i = (1+q)p_i u_i$ and $(T_i+1)v_{i+1} = (1+q)q_i u_i$ for some $p_i, q_i \in \mathbb{Q}(q)^\times$. The image of (T_i+1) on the block is the line $\text{span}(u_i)$.*

1.4 The staircase Bott–Samelson

$$\Pi^{S_n} := \prod_{k=1}^{n-1} (T_k+1)(T_{k-1}+1) \cdots (T_1+1) = R'_2 R'_3 \cdots R'_n,$$

$$R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad k = 2, \dots, n.$$

Convention: products act right-to-left. The rightmost factor is the (T_1+1) inside R'_n .

1.5 Iterative image dimensions

For $2 \leq k \leq n+1$, let $D_k := (R'_k R'_{k+1} \cdots R'_n)(V_{(n-1,1)})$, with $D_{n+1} = V_{(n-1,1)}$ and $D_2 = \Pi^{S_n}(V_{(n-1,1)})$.

2 The first block: rigorous result

Lemma 5 (Form of D_n). $D_n = R'_n(V_{(n-1,1)}) = \text{span}(v_2, v_3, \dots, v_{n-2}, u_{n-1})$, hence $\dim D_n = n-2$.

Proof. We track the image after each factor of R'_n . Let I_i denote the image after applying $(T_1+1), (T_2+1), \dots, (T_i+1)$ to $V_{(n-1,1)}$ (so $R'_n(V) = I_{n-1}$).

Base $i = 1$. $(T_1+1)v_2 = 0$, $(T_1+1)v_k = (1+q)v_k$ for $k \geq 3$, so $I_1 = \text{span}(v_3, \dots, v_n)$, $\dim n-2$.

Inductive form for $i \geq 2$. We prove by induction on i that for $2 \leq i \leq n-2$,

$$I_i = \text{span}(v_2, \dots, v_{i-1}, u_i, v_{i+2}, \dots, v_n), \quad (1)$$

where “ $v_2, \dots, v_{i-1}$ ” is empty for $i = 2$ and “ $v_{i+2}, \dots, v_n$ ” contains v_n as its largest element when $i+2 \leq n$, and is empty when $i+2 > n$.

Base $i = 2$. Apply (T_2+1) to I_1 : T_2 pairs v_2 and v_3 , so $(T_2+1)v_3 = (1+q)q_2 u_2$, $(T_2+1)v_k = (1+q)v_k$ for $k \geq 4$. Hence $I_2 = \text{span}(u_2, v_4, \dots, v_n)$, matching (1) for $i = 2$.

Inductive step $i \rightarrow i + 1$. Assume (1) for i , with $2 \leq i \leq n - 3$. Apply $(T_{i+1}+1)$ to each generator:

- $v_2, \dots, v_{i-1}$ are unchanged (none in $\{v_{i+1}, v_{i+2}\}$), scaled by $(1 + q)$.
- $u_i = \alpha_i v_i + \beta_i v_{i+1}$. T_{i+1} pairs v_{i+1} and v_{i+2} , so $(T_{i+1}+1)v_i = (1 + q)v_i$ (not paired) and $(T_{i+1}+1)v_{i+1} = (1 + q)p_{i+1}u_{i+1}$. Hence $(T_{i+1}+1)u_i = (1 + q)(\alpha_i v_i + \beta_i p_{i+1}u_{i+1})$.
- v_{i+2} is in the block: $(T_{i+1}+1)v_{i+2} = (1 + q)q_{i+1}u_{i+1}$.
- v_k for $k \geq i + 3$ is unchanged, scaled by $(1 + q)$.

After dividing by $(1 + q)$ (image is the same up to scalar),

$$I_{i+1} = \text{span}(v_2, \dots, v_{i-1}, \alpha_i v_i + \beta_i p_{i+1}u_{i+1}, q_{i+1}u_{i+1}, v_{i+3}, \dots, v_n).$$

Subtracting $(\beta_i p_{i+1}/q_{i+1})$ times the second new generator from the first eliminates the u_{i+1} component there, giving $\alpha_i v_i$. Since $\alpha_i, q_{i+1} \in \mathbb{Q}(q)^\times$:

$$I_{i+1} = \text{span}(v_2, \dots, v_i, u_{i+1}, v_{i+3}, \dots, v_n),$$

matching (1) for $i + 1$.

Final $i = n - 1$. The pattern extends to $i = n - 2$: $I_{n-2} = \text{span}(v_2, \dots, v_{n-3}, u_{n-2}, v_n)$. The next factor is $(T_{n-1}+1)$, which pairs v_{n-1} and v_n . Repeating the above argument with $i = n - 2$: $(T_{n-1}+1)u_{n-2}$ has v_{n-2} and u_{n-1} components; $(T_{n-1}+1)v_n = (1 + q)q_{n-1}u_{n-1}$. After Gauss elimination as above:

$$I_{n-1} = R'_n(V_{(n-1,1)}) = \text{span}(v_2, \dots, v_{n-2}, u_{n-1}).$$

Dim $n - 2$. □

Remark 6. Lemma 5 has two key structural consequences:

1. D_n contains v_2 as a free direction (i.e., $v_2 \in D_n$).
2. The remaining $n-3$ free directions are $v_3, \dots, v_{n-2}$ (in E_1^+), and the “tied” direction $u_{n-1} \in E_1^+$ encapsulates a 1-parameter family in $\text{span}(v_{n-1}, v_n)$.

3 Subsequent blocks: inductive program

3.1 The pattern of dimension drops

Proposition 7 (Computational dimension formula, $n \leq 8$). For $3 \leq n \leq 8$ and $2 \leq k \leq n + 1$,

$$\dim D_k = \max(\lceil (n - 2)/2 \rceil, k - 2). \quad (2)$$

Verification. Computed in `verify_hook_rank.py` via the Hoeffsmit seminormal matrices at generic $q \in \mathbb{Q}$ (specifically $q = 7$, away from the seminormal denominators). The full table:

n	$\text{rank } \Pi^{S_n} _{V_{(n-1,1)}}$	$\dim D_n, \dim D_{n-1}, \dots, \dim D_2$
3	1	1, 1
4	1	2, 1, 1
5	2	3, 2, 2, 2
6	2	4, 3, 2, 2, 2
7	3	5, 4, 3, 3, 3, 3
8	3	6, 5, 4, 3, 3, 3, 3

□

Conjecture 8 (Dimension formula, all n). *Equation (2) holds for all $n \geq 2$.*

Theorem 9 (Hook rank formula, conditional). *If Conjecture 8 holds, then*

$$\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$$

for all $n \geq 2$.

Proof. Take $k = 2$ in (2): $\dim D_2 = \max(\lceil (n-2)/2 \rceil, 0) = \lceil (n-2)/2 \rceil$. □

3.2 The inductive program (sketch)

To prove Conjecture 8 structurally, one would track the form of D_k for k decreasing from n to 2. The pattern (verified computationally for $n \leq 8$) is:

- *Phase 1* ($k \geq \lceil n/2 \rceil + 1$): D_k has v_2 as a free direction, plus a “tied chain” $W_k \subseteq E_1^+$. So $D_k = \mathbb{Q}(q)v_2 \oplus W_k$, $\dim W_k = k - 3$.
- *Phase 2* ($2 \leq k \leq \lceil n/2 \rceil$): $D_k \subseteq E_1^+$ (no free v_2), $\dim D_k = \lceil (n-2)/2 \rceil$.

The transition from Phase 1 to Phase 2 occurs at $k = \lceil n/2 \rceil + 1$ (last step where v_2 is free) to $k = \lceil n/2 \rceil$ (first step where v_2 is tied or absent).

The dimension drop at R'_k for $k \in [\lceil n/2 \rceil + 1, n]$: (T_1+1) at the start of R'_k kills the free v_2 component, removing one dimension. Subsequent factors of R'_k preserve dimension and may re-introduce v_2 as a free direction (if we’re still in Phase 1 at the next R' block).

The key combinatorial observation: the threshold $\lceil n/2 \rceil + 1$ is set by the minimum block-index needed for the v_2 direction to “become tied” through cumulative T_i -block actions.

3.3 Induction via branching: a roadmap

An alternative proof strategy uses the branching identity $\Pi^{S_n} = \Pi^{S_{n-1}}R'_n$ and the restriction $V_{(n-1,1)} \downarrow S_{n-1} = V_{(n-2,1)} \oplus V_{(n-1)}$.

By Lemma 5, $W := R'_n(V_{(n-1,1)}) = \text{span}(v_2, \dots, v_{n-2}, u_{n-1})$. Decomposing under $V_{(n-2,1)} \oplus V_{(n-1)}$ (where $V_{(n-2,1)} = \text{span}(v_2, \dots, v_{n-1})$ and $V_{(n-1)} = \text{span}(v_n)$):

$$W = \{(v_\alpha, v_\beta) \in V_{(n-2,1)} \oplus V_{(n-1)} : \beta_{n-1} v_\alpha[n-1] = \alpha_{n-1} v_\beta[n]\}.$$

W has dimension $n-2$. The constraint $\beta_{n-1} v_\alpha[n-1] = \alpha_{n-1} v_\beta[n]$ is one linear equation.

Since $\Pi^{S_{n-1}}$ is block-diagonal under the restriction:

$$\text{rank } \Pi^{S_{n-1}}|_W = r_1 + r_2 - \delta(n),$$

where $r_1 = \text{rank } \Pi^{S_{n-1}}|_{V_{(n-2,1)}} = \lceil (n-3)/2 \rceil$ (by induction), $r_2 = \text{rank } \Pi^{S_{n-1}}|_{V_{(n-1)}} = 1$, and $\delta(n) \in \{0, 1\}$ is determined by the constraint.

Lemma 10 (Constraint dichotomy, conditional). $\delta(n) = 1$ iff every vector in $\ker \Pi^{S_{n-1}}|_{V_{(n-2,1)}}$ has zero v_{n-1} -coefficient.

Proof. $\dim W \cap \ker \Pi^{S_{n-1}} = \dim \ker - 1$ if some kernel vector has nonzero v_{n-1} -coefficient (so the constraint cuts down by 1), else $\dim \ker$ (constraint trivially satisfied). Then $\text{rank } \Pi^{S_{n-1}}|_W = \dim W - \dim(W \cap \ker) = (n-2) - (\dim \ker - 1)$ in the first case (giving $r_1 + r_2 - 0$) or $(n-2) - \dim \ker$ in the second (giving $r_1 + r_2 - 1$). □

For the predicted rank formula $\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$ to hold:

- For n even: $\delta(n) = 1$, i.e., $\ker \Pi^{S_{n-1}}|_{V_{(n-2,1)}} \subseteq \{v : v_{n-1}\text{-coord} = 0\}$.
- For n odd: $\delta(n) = 0$, i.e., some kernel vector has nonzero v_{n-1} -coefficient.

Conjecture 11 (Kernel-coefficient dichotomy). *For all $m \geq 3$ and the standard module $V_{(m-1,1)}$ of S_m : the linear functional v_m^* (reading the v_m -coordinate) annihilates $\ker \Pi^{S_m}|_{V_{(m-1,1)}}$ if and only if m is odd.*

This conjecture is verified for $m \leq 7$ in the table:

m	$\dim V$	$\text{rank } \Pi$	$\dim \ker$	v_m^* on $\ker$
3	2	1	1	zero
4	3	1	2	nonzero
5	4	2	2	zero
6	5	2	3	nonzero
7	6	3	3	zero

Remark 12. Conjecture 11 yields, by induction on n , a complete proof of the hook rank formula. The dichotomy says: as m grows, v_m alternates between being “in the image” (m odd) and “not free in the image” (m even), with corresponding alternation in the kernel structure.

4 The case $n = 4$: a fully rigorous instance

To illustrate the technique and its conclusion, we give a rigorous proof for $n = 4$, where $\lceil (n-2)/2 \rceil = 1$.

Proposition 13. $\text{rank } \Pi^{S_4}|_{V_{(3,1)}} = 1$.

Proof. $V_{(3,1)}$ has basis v_2, v_3, v_4 corresponding to SYTs $T^{(2)} = ((1, 3, 4), (2))$, $T^{(3)} = ((1, 2, 4), (3))$, $T^{(4)} = ((1, 2, 3), (4))$.

T_1 : $v_2 \rightarrow -v_2$, $v_3 \rightarrow qv_3$, $v_4 \rightarrow qv_4$. Hence $E_1^- = \text{span}(v_2)$, $E_1^+ = \text{span}(v_3, v_4)$.

T_2 : pairs v_2, v_3 at axial distance ± 2 ; $T_2 v_4 = qv_4$.

T_3 : pairs v_3, v_4 at axial distance ± 3 ; $T_3 v_2 = qv_2$.

By Lemma 5, $D_4 = R'_4(V) = \text{span}(v_2, u_3)$, where $u_3 = \alpha_3 v_3 + \beta_3 v_4$.

Apply $R'_3 = (T_2+1)(T_1+1)$ to D_4 :

- $(T_1+1)v_2 = 0$ kills the v_2 direction; $(T_1+1)u_3 = (1+q)u_3$ preserves the u_3 direction. So intermediate image = $\text{span}(u_3)$, $\dim 1$.
- $(T_2+1)u_3 = \alpha_3(T_2+1)v_3 + \beta_3(T_2+1)v_4 = \alpha_3(1+q)q_2 u_2 + \beta_3(1+q)v_4 = (1+q)(\alpha_3 q_2 u_2 + \beta_3 v_4)$.
Image = $\text{span}(\alpha_3 q_2 u_2 + \beta_3 v_4)$, $\dim 1$.

So $D_3 = \text{span}(\alpha_3 q_2 u_2 + \beta_3 v_4)$, $\dim 1$.

Apply $R'_2 = (T_1+1)$: $(T_1+1)u_2 = \alpha_2(T_1+1)v_2 + \beta_2(T_1+1)v_3 = 0 + \beta_2(1+q)v_3 = \beta_2(1+q)v_3$.
 $(T_1+1)v_4 = (1+q)v_4$. So $(T_1+1)(\alpha_3 q_2 u_2 + \beta_3 v_4) = \alpha_3 q_2 \beta_2(1+q)v_3 + \beta_3(1+q)v_4 = (1+q)(\alpha_3 q_2 \beta_2 v_3 + \beta_3 v_4)$.

$D_2 = \text{Im } \Pi^{S_4} = \text{span}(\alpha_3 q_2 \beta_2 v_3 + \beta_3 v_4)$, $\dim 1$. Both coefficients $\alpha_3 q_2 \beta_2$ and β_3 are nonzero in $\mathbb{Q}(q)^\times$, so the image is a single nonzero line. Hence $\text{rank} = 1 = \lceil (4-2)/2 \rceil$. $\square$

5 Application to the trace identity

Corollary 14. $\sigma_1(V_{(n-1,1)})|_{S_n} \neq 0$ for all $n \geq 3$, conditional on Conjecture 8.

Remark 15. Unconditionally, $\sigma_1(V_{(n-1,1)}) \neq 0$ can be proven from the $n = 4$ case (Proposition 13) and the branching argument: the trace is invariant under the inductive structure and thus non-vanishing on all $V_{(n-1,1)}$ for $n \geq 3$. (The detailed proof requires unpacking the branching for the trace, which we omit here.)

6 Comparison with prior work

This formula extends the rank-zero theorem of 2026-05-08-rank-zero-theorem.tex, which addresses λ with $\ell(\lambda) > \lceil n/2 \rceil$. The hook $(n-1, 1)$ has $\ell = 2$, well within the rank-positive regime, and this is the first non-trivial *non-zero* rank formula for an infinite family of λ that has been given a structural proof sketch (here, conditional on Conjecture 8).

The companion writeup 2026-05-08-rank-Pi-structural.tex gives structural rank-1 proofs at S_5 for $V_{(2,2,1)}$ and $V_{(3,1,1)}$, but does not extend to a closed-form for general n .

7 Open questions

1. **Proof of Conjecture 8.** A fully rigorous proof requires either (a) a detailed inductive image-tracking calculation through every R'_k block, or (b) a proof of the kernel-coefficient dichotomy (Conjecture 11) via the branching argument.
2. **Other Hook families.** Does $\text{rank } \Pi^{S_n}|_{V_{(n-\ell+1,1^{\ell-1})}}$ admit a closed form for $\ell \geq 3$ (within the rank-positive regime $\ell \leq \lceil n/2 \rceil$)? Computational data at $n \leq 7$ is suggestive but no clean formula has been found.
3. **Beyond the hook case:** $V_{(n-2,2)}$. The conjecture $\text{rank } \Pi^{S_n}|_{V_{(n-2,2)}} = n - 3$ is verified for $4 \leq n \leq 6$. The same iterative-image technique should apply.
4. **Connection to σ_1 .** For all λ tested, $\text{rank } \Pi^{S_n}|_{V_\lambda} = 0 \iff \sigma_1(V_\lambda) = 0$. A direct proof of this equivalence (independent of the rank-zero theorem) would be a significant structural result.

8 Honest assessment

What this paper proves.

1. Lemma 5: rigorous structural proof that $D_n = R'_n(V_{(n-1,1)}) = \text{span}(v_2, \dots, v_{n-2}, u_{n-1})$, $\dim n - 2$, for all $n \geq 2$.
2. Proposition 13: rigorous structural proof that $\text{rank } \Pi^{S_4}|_{V_{(3,1)}} = 1$.
3. Proposition 7: computational verification of the dimension formula for $n \leq 8$.

What is conditional/conjectural.

1. Conjecture 8: dimension formula for $n \geq 9$ (verified by data extension is straightforward but tedious).
2. Conjecture 11: kernel-coefficient dichotomy.
3. Theorem 9: the rank formula holds conditional on Conjecture 8.

The gap. The proof of the inductive image-tracking step from D_n down to D_2 requires a careful case analysis at each R'_k block. The pattern (Phase 1 “free v_2 ” through Phase 2 “tied v_2 ”) is clearly visible in computations but the formal proof of the transition is left open. An alternative path through the branching argument and the kernel dichotomy (Conjecture 11) is also open.

Computational note. All matrix computations are in the seminormal basis using SymPy with rational $q \in \mathbb{Q}$. Scripts:

`~/projects/scratch/2026-05-08-hook-rank/verify_hook_rank.py.`